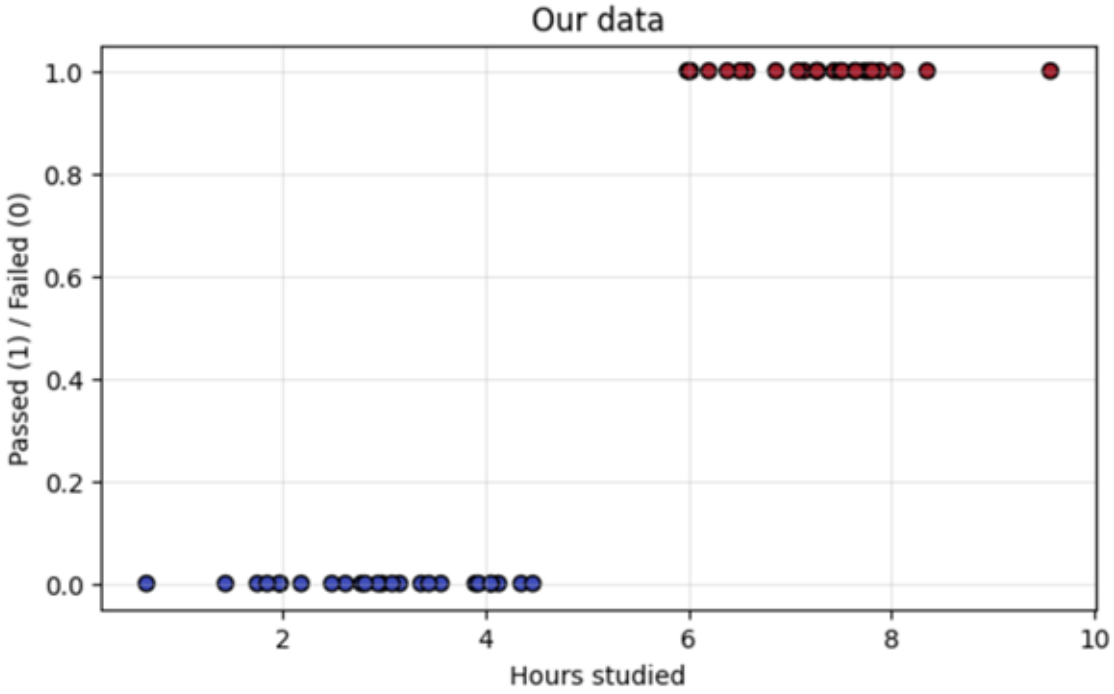
Logistička regresija

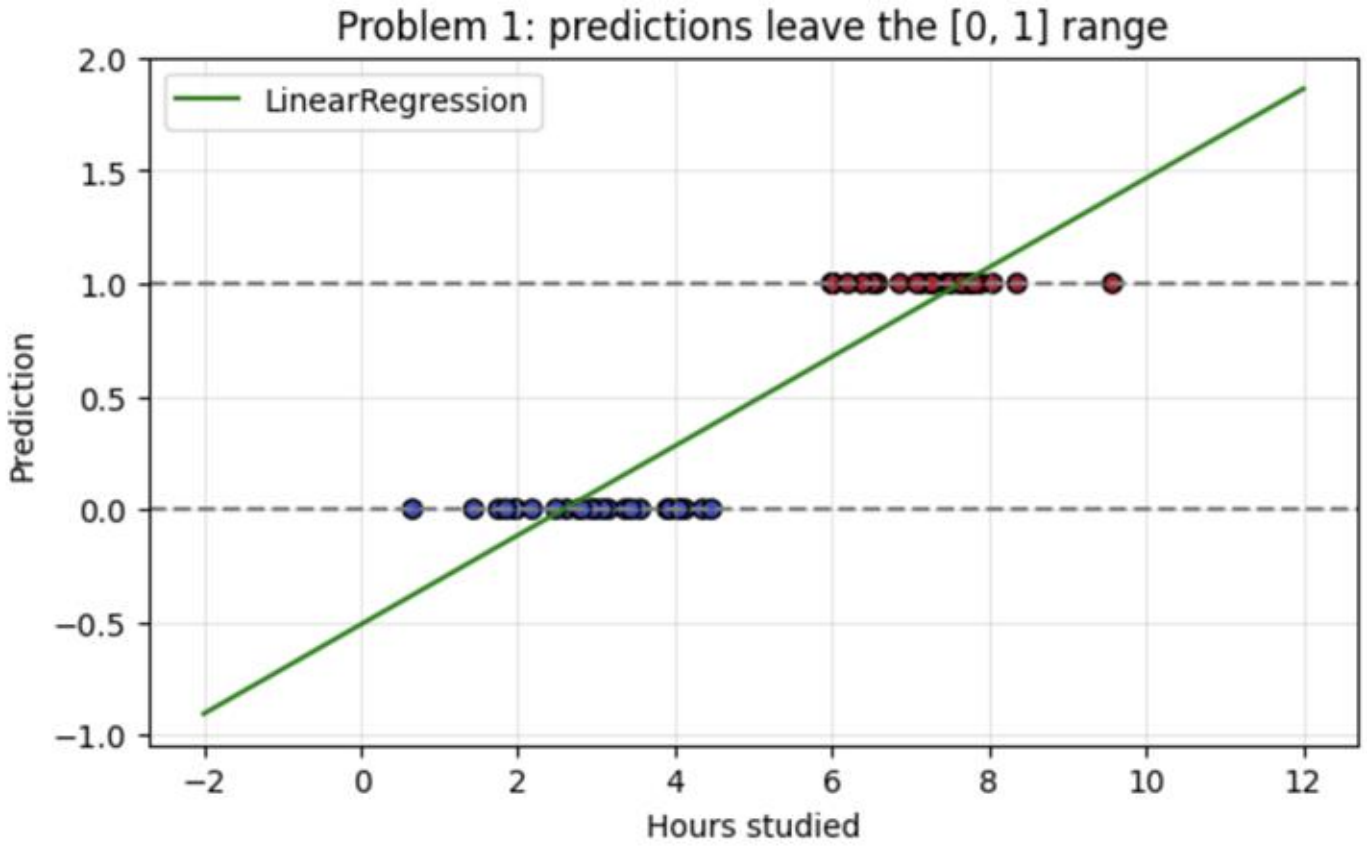
Jelena Jovanovic

Napomena: sve slike na slajdovima su kreirane koriscenjem Claude-a

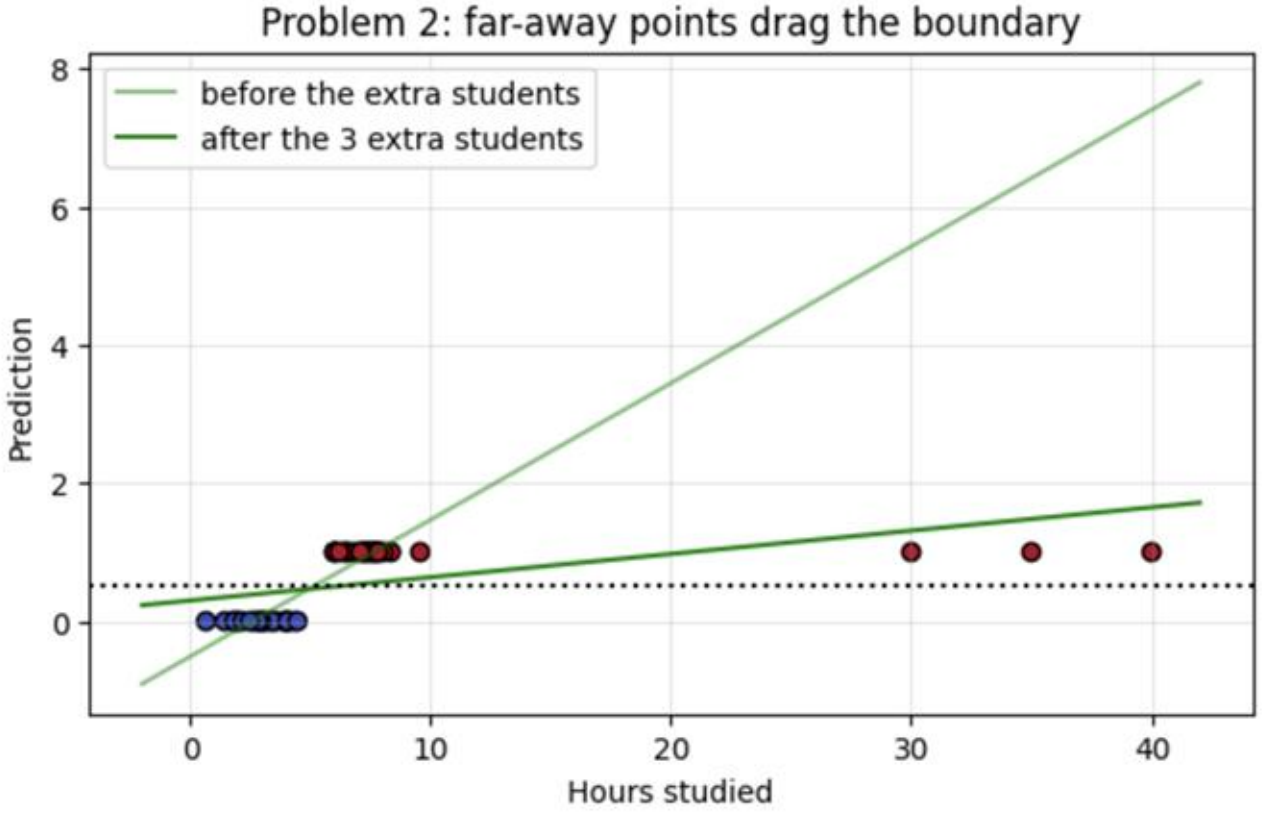
Primer: predviđanje uspeha studenta na ispitu (položio / pao) na osnovu broja sati pripreme



Predviđanje primenom linearne regresije

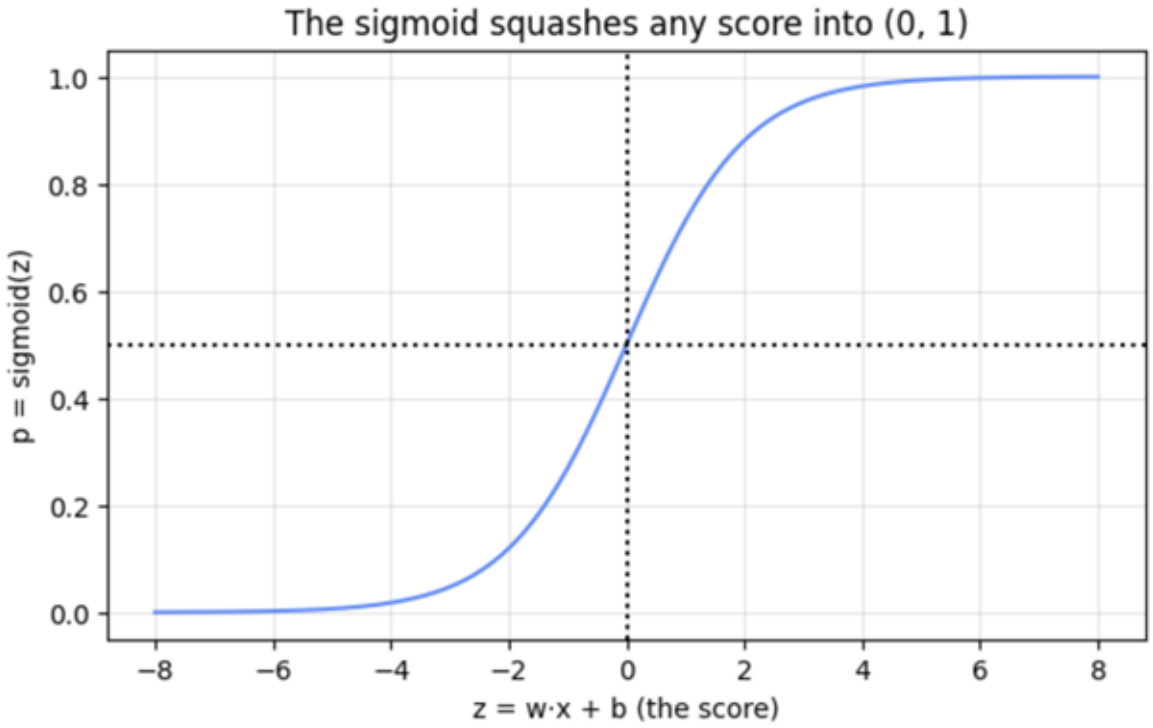


Predviđanje primenom linearne regresije



Granica predviđanja
> pre: 5.11 h
> posle : 5.96 h

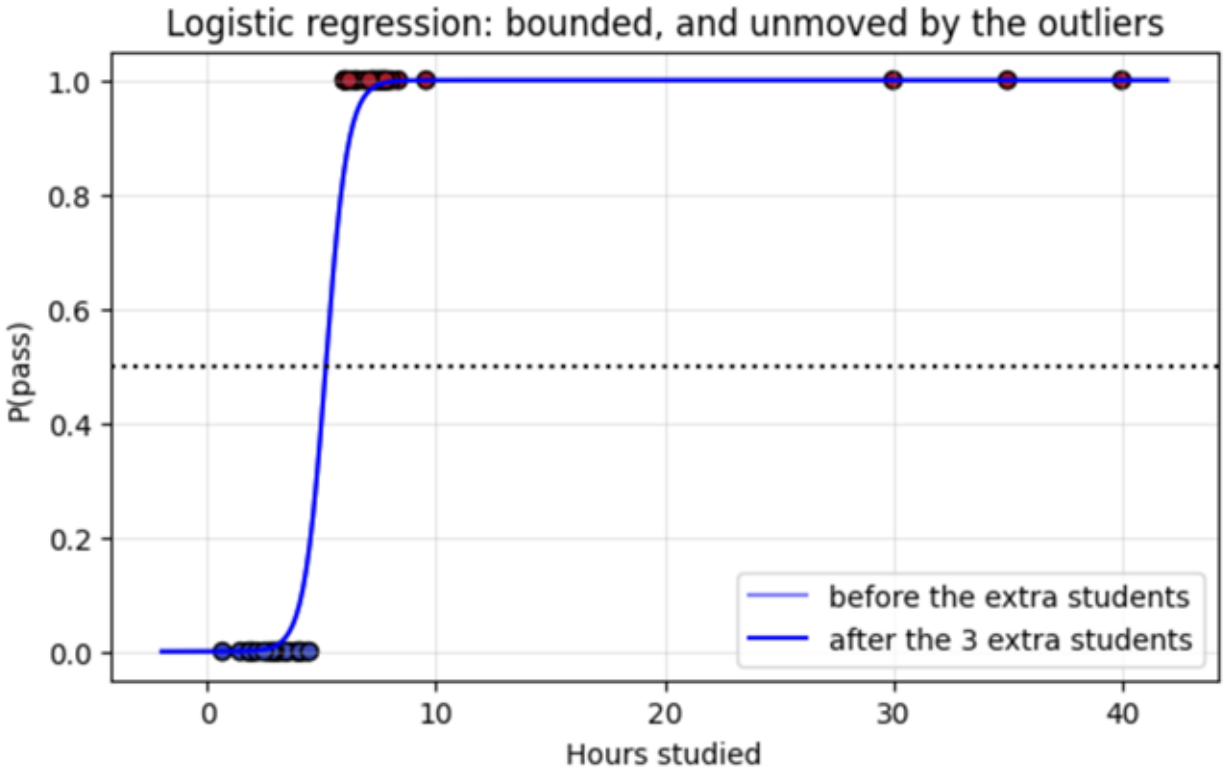
Uvođenje sigmoidne funkcije



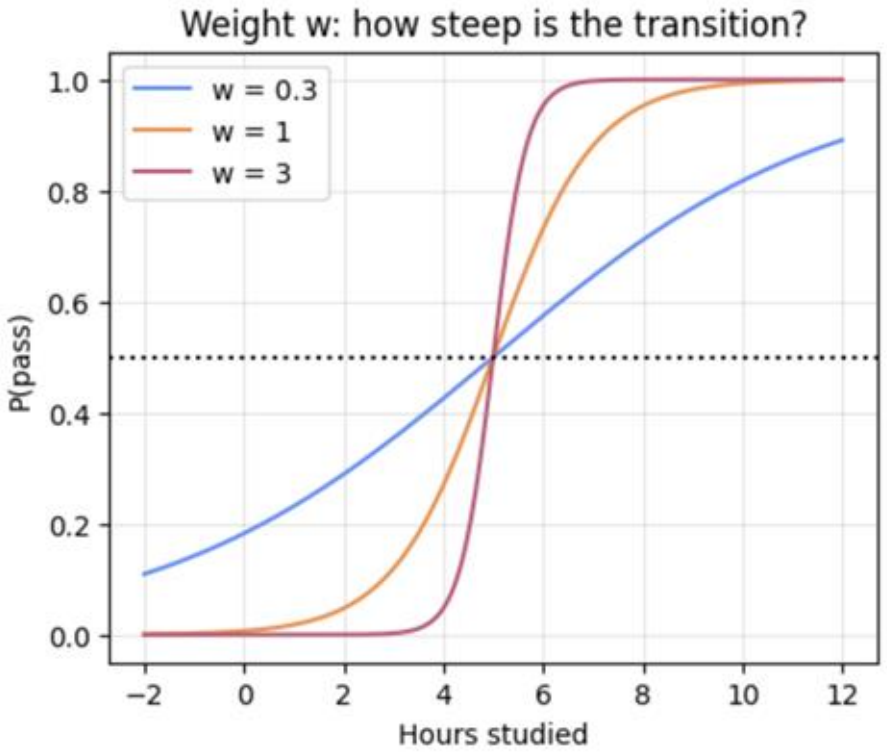
$$z = b + w \cdot x$$

$$p = \sigma(z) = \frac{1}{1 + e^{-z}}$$

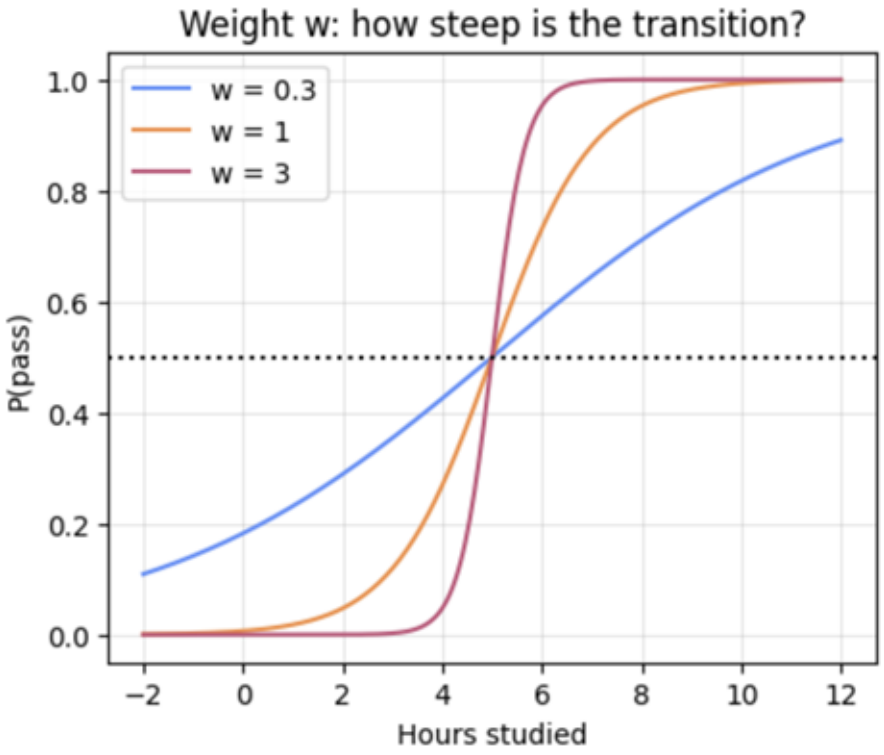
Primena logističke regresije na polaznom primeru

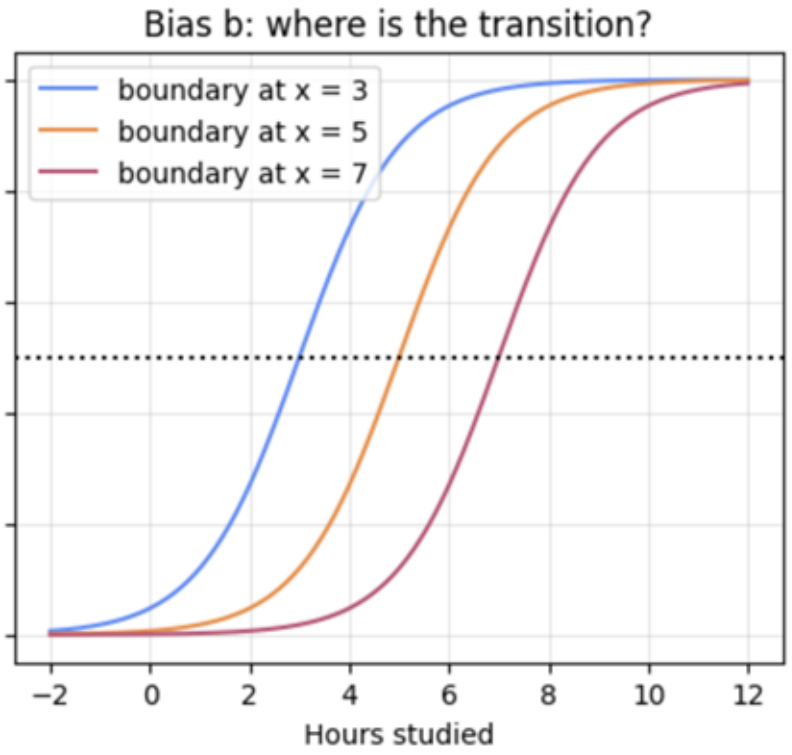


Uticaj parametara na sigurnost modela i granicu odluke

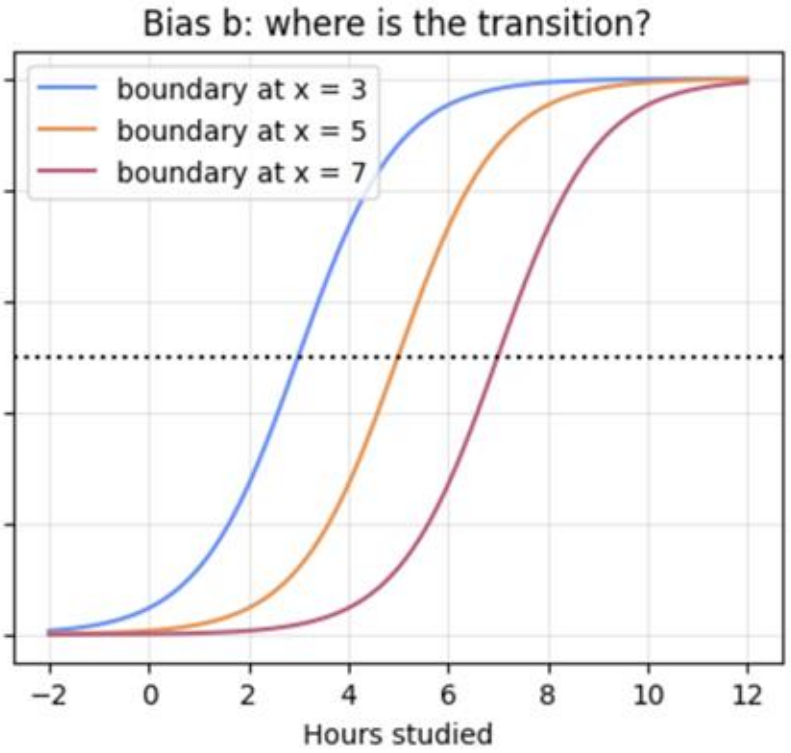


Uticaj parametara na sigurnost modela i granicu odluke





Uticaj parametara na sigurnost modela i granicu odluke



Granica predviđanja: $z = 0$

$$z = b + w \cdot x$$

$$x = -b/w$$

Regularizacija i hiperparametar C

C	Mala vrednost	Visoka vrednost
Regularizacija	Jaka	Slaba
Težine	Teže nuli	Slobodno rastu
Sigmoid	Ravna, “oprezna”	Strma, “sigurna”
Rizik	underfitting	overfitting

Regularizacija i hiperparametar C

